



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID Bridge Program Physics

Lesson 6 Uncertainties and errors

Vocabulary



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Significant figure 有效数字

Decimal place (几位) 小数

Multiplication 乘法

Division 除法

Subtraction 减法

Round (a number) 取近似数

Instrument 器材

Multiple 多次的

Propagation 传播

Significant Figures



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- The number of significant figures indicates how precise a measurement/reading/number is
- Some examples

Number	s.f.
324	3
32400	3
3240	3
3204	4
3204.004	7
32004	5
0.0324	3
0.32400	5
3.2400×10^7	5

How many s.f. does a number have?

Look at the example.

Can you conclude a few rules?

Significant Figures



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- A digit that is not a zero will always be significant
- Zeros that occur sandwiched between non-zero digits are always significant
- Non-sandwiched zeros that occur to the left of a non-zero digit are not significant
- Zeros that occur to the right of the decimal point are significant, providing it's to the right of a non-zero number
- For a number expressed in scientific notation, its s.f. is the number of digits in coefficient

(Bear in mind this is just one of many ways to express the rules of s.f. As long as it works!)

Rounding your answers



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- In multiplication or division (or in raising a number to a power or taking a root), the result must have as many significant figures as the least precisely known number entering the calculation
- In adding and subtracting, the number of decimal places in the answer must be equal to the least number of decimal places in the numbers added or subtracted.



Systematic and Random errors

- Physics is an experimental science – we take measures
- Will any measure be completely accurate? Why or why not?
- Types of errors
 - Systematic errors
 - Always too large or too small
 - Often associated to the measuring instrument (not properly zeroed or calibrated), or measuring techniques
 - In theory, predictable; in real experiment, hard to find
 - When systematic errors are small, the measure is accurate
 - Random errors/random uncertainties
 - repeated measurements of the same quantity show a spread of values, some too large some too small (unbiased)
 - Averaging a large number of measurements gives a more accurate estimate of the result

Precision and accuracy



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High accuracy
High precision



Low accuracy
High precision



High accuracy
Low precision



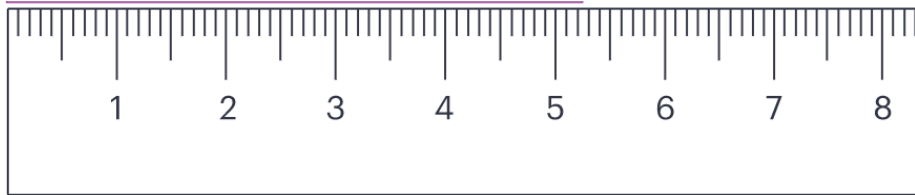
Low: accuracy
Low: precision

The uncertainty of a Measurement



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- Uncertainty is partially determined by the measuring instrument:



- Analogue instrument (e.g. a ruler): half of the smallest division of the instrument. E.g. when you use the ruler above, uncertainty is ± 0.05 cm
 - Digital display instrument (e.g. a digital scale): the smallest division that the instrument can read. The uncertainty given by the scale above is ± 0.1 kg
- Uncertainty also depends on the observer.

The uncertainty of multiple measurements



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- In an experiment a measurement can be repeated many times
- If it is repeated N times and the results of the measurements are $x_1, x_2 \dots \dots x_n$, we calculate the mean or the average of these values $\bar{x}$

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

- A very simple way (though not the only way) to get uncertainty of the data Δx is

$$\Delta x = \frac{x_{\max} - x_{\min}}{2}$$

The way we show uncertainty



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For a length $L = (28.3 \pm 0.4) \text{ cm}$

- absolute uncertainty = 0.4 cm
- fractional uncertainty = $\frac{0.4}{28.3} = 0.0141$
- percentage uncertainty = $0.0141 \times 100\% = 1.41\%$

Uncertainties in calculation (propagation of uncertainty)



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Addition and subtraction

$$Q = a + b \quad \Rightarrow \quad \Delta Q = \Delta a + \Delta b$$

$$Q = a - b \quad \Rightarrow \quad \Delta Q = \Delta a + \Delta b$$

$$Q = a + b - c \quad \Rightarrow \quad \Delta Q = \Delta a + \Delta b + \Delta c$$

Multiplication and division

$$Q = ab \quad \Rightarrow \quad \frac{\Delta Q}{Q_0} = \frac{\Delta a}{a_0} + \frac{\Delta b}{b_0}$$

$$Q = \frac{a}{b} \quad \Rightarrow \quad \frac{\Delta Q}{Q_0} = \frac{\Delta a}{a_0} + \frac{\Delta b}{b_0}$$

$$Q = \frac{ab}{c} \quad \Rightarrow \quad \frac{\Delta Q}{Q_0} = \frac{\Delta a}{a_0} + \frac{\Delta b}{b_0} + \frac{\Delta c}{c_0}$$

Powers and roots

$$Q = a^n \quad \Rightarrow \quad \frac{\Delta Q}{Q_0} = |n| \frac{\Delta a}{a_0}$$

$$Q = \sqrt[n]{a} \quad \Rightarrow \quad \frac{\Delta Q}{Q_0} = \frac{1}{n} \frac{\Delta a}{a_0}$$

There shouldn't be any minus sign when working out the uncertainty in a calculated figure



You don't get more precise numbers just by doing some calculation on them!



Uncertainty propagation example

Find the percentage uncertainty in the quantity $Q = a - b$, where $a = 538.7 \pm 0.3$ and $b = 537.3 \pm 0.5$

$$Q_o = a_o - b_o = 1.4$$

The absolute uncertainty $\Delta Q = 0.3 + 0.5 = 0.8$.

$$\text{So } Q = 1.4 \pm 0.8$$

*The fractional uncertainty is $\frac{0.8}{1.4} = 0.57$
(percentage uncertainty is 57%)*

The fractional uncertainty in the quantities a and b is quite small. But the numbers are close to each other so their difference is very small. This makes the fractional uncertainty in the difference unacceptably large.



Uncertainty propagation example

The sides of a rectangle are measured to be $a = 2.5 \text{ cm} \pm 0.1 \text{ cm}$ and $b = 5.0 \text{ cm} \pm 0.1 \text{ cm}$. Find the area A of the rectangle.

The fractional uncertainty in a is:

$$\frac{\Delta a}{a} = \frac{0.1}{2.5} = 0.04 \text{ or } 4\%$$

The fractional uncertainty in b is:

$$\frac{\Delta b}{b} = \frac{0.1}{5.0} = 0.02 \text{ or } 2\%$$

Thus, the fractional uncertainty in the area is $0.04 + 0.02 = 0.06$ or 6%.

The area A_0 is:

$$A_0 = 2.5 \times 5.0 = 12.5 \text{ cm}^2$$

and $\frac{\Delta A}{A_0} = 0.06$

$$\Rightarrow \Delta A = 0.06 \times 12.5 = 0.75 \text{ cm}^2$$

Hence $A = 12.5 \text{ cm}^2 \pm 0.8 \text{ cm}^2$ (the final absolute uncertainty is quoted to 1 s.f.).

- The final absolute uncertainty is rounded to 1 significant figure (e.g. 0.8)
- The final result of the area (12.5) need to keep the same decimal place with the uncertainty part.